

Inventory-aware design: the latest FlowPilot addition

A laboratory profile is built once, versioned, and enforced in every batch-to-flow translation.

1

Collect

Accept laboratory PDFs, PPTX/DOCX files, pasted JSON, or structured forms.

2

Normalize + freeze

Convert equipment and operating constraints into one reproducible profile (v2.0).

3

Enforce

Snap designs to listed hardware and block unresolved reactor trains, mixers, BPRs, or limits.

Example: KHU laboratory profile

khu_laboratory_inventory v1 | schema v2.0

7

Reactors

3

Pumps

3

Tubing

2

Gas hardware

0

BPR settings

HARD CONSTRAINT

No inline degasser | BPR not confirmed

Design consequence The agent may propose chemistry, but only inventory-compatible parameters can be shown as an executable design.

Inventory management: end-to-end workflow

The profile is generated independently, then injected as a frozen hard-constraint layer before design execution.

